Design and Evaluation of a myDAQ-Controlled, Stepper-Driven LiDAR Mälardalen University

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***Abstract*—This work presents a motor-controlled LiDAR using an NI myDAQ, a double H-bridge stepper drive, and a configurable sweep (default 180 degrees, 1.8 degree steps). The sensor was calibrated to $D\,[\mathrm{cm}] = 27.16\,V^{-0.714}$. In RQ1, the smallest reliably detectable object increased with distance: 12×19 mm was not reliable at 10-20 cm, while about 56×80 mm was required at 80 cm. In RQ2, the minimum undetectable speed rose from roughly 0.2-0.3 m/s at 10-20 cm to about 0.7 m/s at 80 cm. In RQ3, matte, opaque targets were detected more consistently than glossy or transparent ones. Measurements were taken in a room with daylight, and a PCB failure reduced repeated trials. Despite these constraints, the system produced stable scans and credible bounds for size and motion detection, and we outline straightforward improvements to increase repeatability and range.**

***Index Terms*—H-bridge, Stepper Motor, LiDAR, PCB, Multisim, Ultiboard, LabVIEW, myDAQ**

I. Introduction

In the rapidly advancing field of robotics and automated systems, the integration of LiDAR (Light Detection and Ranging) technology has become essential for accurate mapping and distance measurement. This project focuses on the construction and analysis of a motor-controlled LiDAR system, using a stepper motor for sensor positioning, and the construction of a double H-bridge circuit PCB that will handle motor control. The primary aim is to utilize the LiDAR system for accurate distance measurement and object detection, which is critical in various applications such as autonomous vehicles, robotics, and environmental monitoring.

Light Detection and Ranging (LiDAR), is a remote sensing technology that utilizes laser light to determine distances and generate precise three-dimensional data about the shape of objects and their surface characteristics [1]. Its functionality shares similarities to radar, which utilizes radio waves; the key difference is that LiDAR utilizes laser light waves [2].

The motor-controlled LiDAR system in this project is designed to capture accurate distance measurements by rotating the LiDAR sensor, thereby covering a wider range of areas and providing a detailed representation of the surroundings. The double H-bridge circuit, consisting of two H-bridges, is employed to control the direction and speed of the stepper motor, ensuring precise positioning of the sensor. This integration of mechanical and electronic components forms the basis of our LiDAR system.

In this project to create a functional motor-controlled LiDAR system we had to utilize a lot of different aspects. Firstly, we utilized Multisim to design and simulate the double H-bridge circuit, a crucial component of our system that allows us to control the direction and speed of the motor. Secondly, the circuit design was exported to Ultiboard, where we meticulously laid out the components and connections on our printed circuit board (PCB). The design was then exported to FlatCAM, a CNC software, which translated our design into G-code for the CNC machine to cut out the PCB accurately. Finally, to control the motor and process the LiDAR data, we utilized LabVIEW, a system-design platform and development environment.

II. Problem Formulation

The primary objective of this project was to successfully design and implement a motor-controlled LIDAR system. Concurrently, the main focus of this research paper revolves around conducting an in-depth investigation to address the following three research questions and answer them appropriately:

RQ1: What is the smallest object that the radar can detect depending on distance from the sensor?

RQ2: What is the minimum speed the object can move without being detected?

RQ3: What physical properties of the object make it harder or easier to be detected?

These questions will be investigated through both theoretical analysis and empirical measurements, with the goal of understanding the limitations and potential improvements for our LiDAR system.

III. Method

To carry out this project, a systematic methodology was followed to develop the motor-controlled LiDAR system. The first step was to simulate the behaviour of a double H-bridge that was capable of driving forward and backwards. A bipolar stepper requires two full H-bridges—one per winding—to reverse current in each phase for bidirectional control. The simulation aspect was created in NI Multisim, and a visual representation is provided in Figure 2. The simulation was an essential step to identify and thereby rectify any potential issues before the physical assembly.

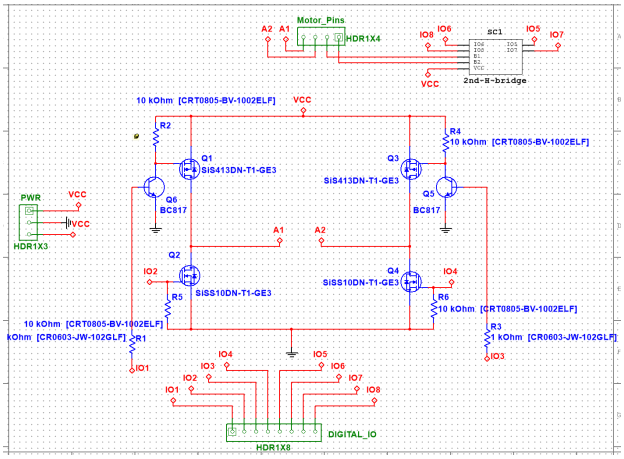


Figure 1. First H-bridge & Connectors

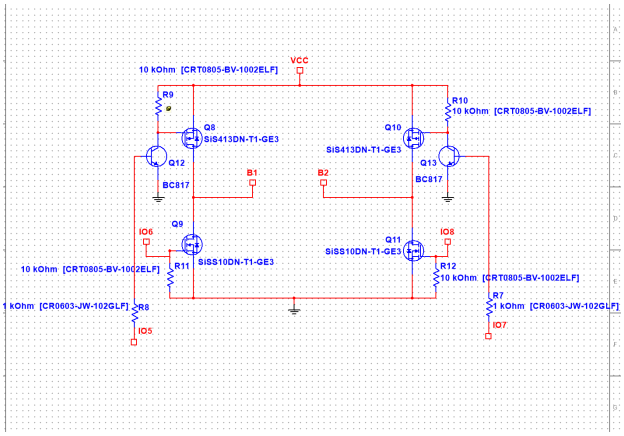


Figure 2. Second H-bridge

Post-simulation, the circuit constructed in Multisim was exported to Ultiboard in order to create a layout for the PCB, whilst also ensuring that all of the components and connections were appropriately represented in the layout. Ultiboard provides a comprehensive environment for circuit board design, allowing the decision on component placements and route connections. Moreover, the next step was to place the components on the board, taking into consideration factors such as the size of the components, the required spacing, and the desired orientation before routing them using the auto-routing function and manual routing. This PCB resulted in a compact one-sided design, as shown in Figure 3. Lastly, configurations of settings and layout was done before exporting this PCB as Gerber files to FlatCAM.

Upon completion of the PCB design, the Gerber files for the isolation, routing, and drill holes were exported to FlatCAM. This step was crucial as it facilitated the transition from a digital design to a physical product.

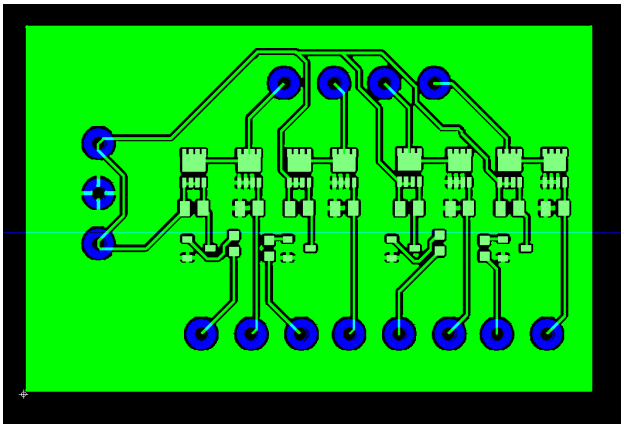


Figure 3. PCB design in Ultiboard.

FlatCAM interprets the Gerber files produced by Ultiboard, translating the design to G-code, which essentially is a series of commands that instruct the CNC machine on how to move, turn, and control the spindle to accurately cut the PCB. This process ensures that the design specifications are precisely followed, resulting in a physical circuit board that accurately represents the original design. A visual representation of the FlatCAM export is in Figure 4.

Upon completion of the milling process, the integrity

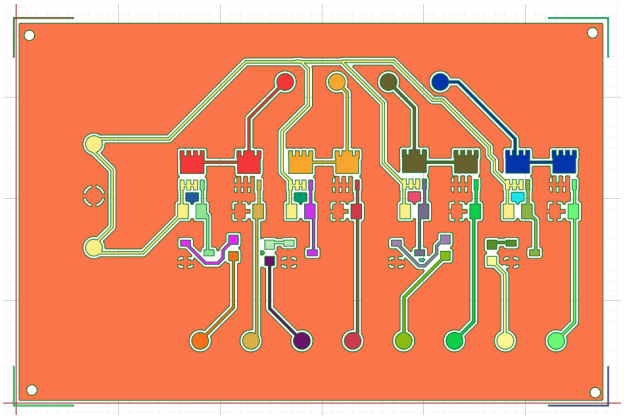


Figure 4. PCB exported to FlatCAM for CNC milling.

of the PCB was verified with a multimeter to ensure proper functionality and continuity. The next step involved soldering the electronic components to their specific locations on the board. A 3D print of the PCB was created in SolidWorks and affixed to the surface of the PCB with the exception of the component placements being cut out in order to manually apply soldering paste. With the solder paste in place and the components positioned, the assembly was done by utilizing an LPKF ProtoFlow E oven [3] for the soldering process to be finalized. The connectors, on the other hand, were soldered manually to the drill holes from the opposite side.

The final step involved developing the motor-control VI

in LabVIEW. The VI interfaces with the NI myDAQ to generate step and direction pulses for the double H-bridge (stepper motor) while, at each step, simultaneously acquiring the LiDAR's analog voltage. The measured voltage is converted to distance using the calibration and plotted. Motor direction is reversed at predefined sweep limits. The mechanism is capable of 360° rotation; in this study the sweep was configured to 180° with fixed angular limits. The overall communication structure is shown in Figure 5, and the stepping/measurement logic in Figure 6.

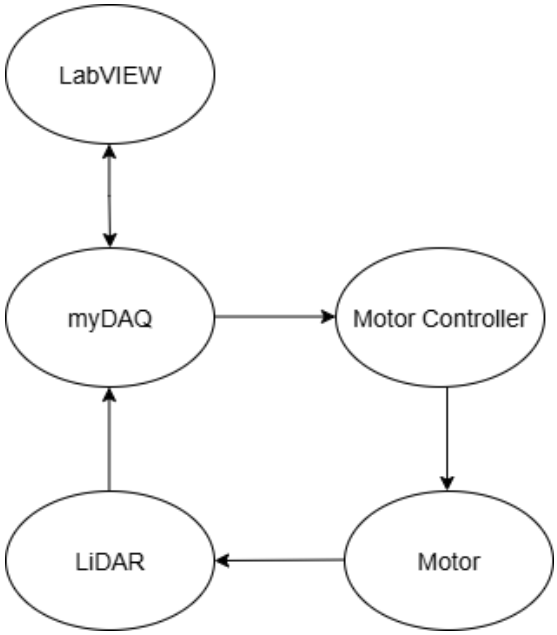


Figure 5. System communication between LabVIEW, myDAQ, motor controller, LiDAR sensor, and motor.

With the system fully assembled, we carried out extensive testing to ensure that everything was functioning as expected. During testing, we carefully observed and recorded the system's response to different inputs, making necessary adjustments to the software and hardware as needed. This comprehensive methodology allowed us to successfully develop a functional and reliable motor-controlled LiDAR system.

IV. Hardware

The hardware utilized in this project played a pivotal role in making the motor-controlled LiDAR system work as intended. This section will explain the functionality of all hardware used and the motives behind using them.

A. myDAQ

For this project, the myDAQ plays a pivotal role in controlling the stepper motor, processing signals, and providing a visual representation of the data acquisition of our LiDAR system. The myDAQ interfaces with the double H-bridge circuit to control the stepper motor. The myDAQ sends signals to the H-bridge, which in turn drives the motor, allowing it to rotate and change the direction of the LiDAR sensor. The myDAQ

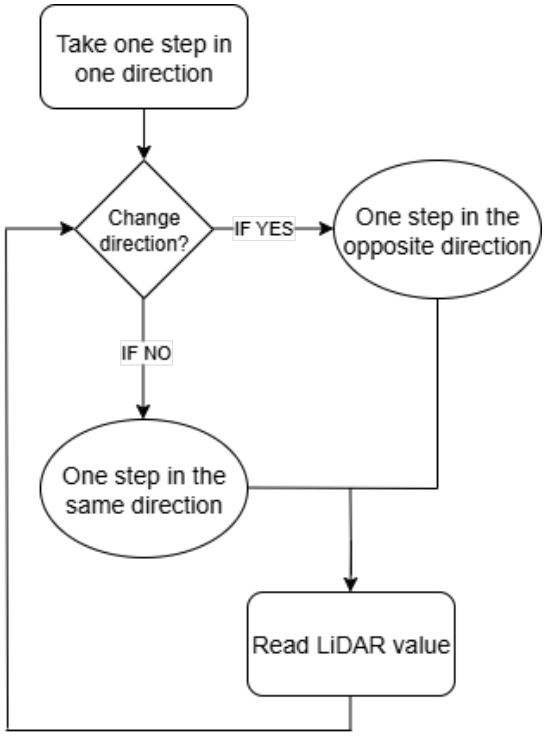


Figure 6. Flowchart of the motor stepping logic and LiDAR measurement process.

acquires signals from the LiDAR sensor. These signals represent the distance measurements obtained by the sensor. The myDAQ processes these signals to extract meaningful data that can be used for analysis. Additionally, the myDAQ is fully compatible with LabVIEW, a graphical programming software, which will be discussed in the Software section.

B. Stepper Motor & LiDAR sensor

In this project, the Sanyo Denki stepper motor [4] was utilized. The selected motor features specifications such as a 1A current rating, 0.39Nm of torque, and a step angle of 1.8 degrees. The stepper motor was a key component for the project due to it enabling the rotation of the mounted LiDAR sensor. This allows it to scan its environment in all degrees (meaning 360 degrees). This motor provides precise control over the positioning of the sensor, which facilitates accurate data collection and mapping of its surroundings. The stepper motor is controlled by the constructed double H-bridge PCB circuit, which enables full control of the direction of the steps of the motor using LabVIEW.The LiDAR sensor used was the SHARP GP2Y0A21YK0F [5].

By integrating the stepper motor with the provided LiDAR sensor and the constructed PCB, the stepper motor facilitated precise and controlled data acquisition for this LiDAR system. The selection of this specific stepper motor and LiDAR sensor was dictated primarily by their accessibility and the adequacy of their characteristics to fulfill the project's requirements.

C. Wegstr CNC Milling Machine

To cut out the PCB, a Wegstr CNC milling machine [6] was utilized. A single-sided fiberglass copper laminate, sized 100mm x 160mm x 1.5mm, was utilized for the construction of the PCB. The isolation routing was created using a 0.2mm, 60-degree spiral bit, which achieved an approximate trace width of 0.3mm and a cut depth (cut-z) of 0.1mm. The outline of the PCB was milled with a 0.2mm, 60-degree spiral bit, operated across 16 iterations utilizing the multi-depth function, thereby milling 0.1mm per iteration.

D. PCB

The PCB hosted the double H-bridge together with supporting components. The switching network was built from BJT-NPN transistors [7], N-channel MOSFETs [8], and P-channel MOSFETs [9], arranged so that the N-MOSFETs handled the low-side switches while the P-MOSFETs controlled the high side. Surface-mounted components were selected to comply with the compact single-sided design required by the Wegstr milling process.

In addition, the PCB included passive components such as four 10 kΩ resistors [10] and four 1 kΩ resistors [11], as well as connectors for signal and motor connections. Since only standard connectors were available, an 8-pin motor connector was fabricated by combining two 4-pin headers [12]. The final PCB was compact and functional, though the single-sided layout made soldering and heat dissipation more challenging.

V. Software

The first step in designing the PCB was to draw up the circuit in NI Multisim and to run simulations to make sure the circuit behaved as desired with the chosen components. The circuit consists of two H-bridges working together as a double H-bridge. The reason for using a double H-bridge is to be able to run the Stepper Motor both forward and backward, since having only one H-bridge only allows for running in one direction. The circuit diagrams can be seen in Figures 1 and 2.

Once the circuit was completed in Multisim, it was exported to NI Ultiboard. When exporting the Multisim files to Ultiboard all the components are brought over to Ultiboard, where they have to be placed according to various PCB design guidelines. The finished Ultiboard design can be seen in Figure 3.

When the PCB design was completed in Ultiboard, it was exported to FlatCAM in order to produce CNC code that could be sent to a PCB CNC machine. The CNC machine used in this project is made by Wegstr; they also make their own CNC software used to control the machine, which was used.

The control system was developed in LabVIEW, which served as the interface between the myDAQ and the hardware. The program generated digital signals to drive the double H-bridge and stepper motor, while simultaneously acquiring the analog voltage from the LiDAR sensor. Each motor step triggered a voltage measurement that was converted into a distance value using the calibration function. These values were then stored and displayed in a .xlsx file. A graphical user interface (GUI) was also implemented, allowing the user to configure parameters such as sweep angle and motor speed.

VI. Results

The completed motor-controlled LiDAR system executed a configurable sweep with a step angle of 1.8°. The default sweep span was 180°, though this could be adjusted in the LabVIEW interface. At each step, the LiDAR provided an analog voltage that was sampled and converted to distance, then the logged data were exported to a `.xlsx` file for analysis. The control chain (myDAQ $\rightarrow$ double H-bridge PCB $\rightarrow$ stepper motor) operated stably and produced continuous measurement sweeps.

The LiDAR's distance response followed a power law of the form

$$D = a\,V^{-b}, \tag{1}$$

where D is the distance in centimeters and V is the sensor output voltage in volts. For this system, the fitted calibration was

$$\boxed{D\ [\text{cm}] = 27.16\,V^{-0.714}} \tag{2}$$

which provided practical accuracy of approximately ± 1 cm up to 50 cm, and $\pm$(2-3) cm from 50 cm to 80 cm. A regression plot (voltage vs. distance) confirmed the expected inverse, nonlinear behavior.

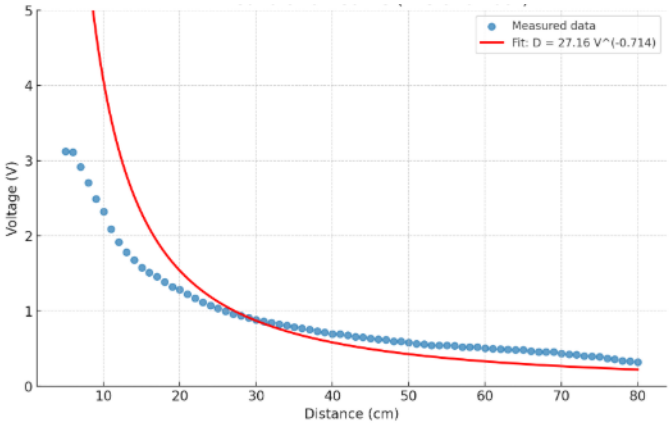


Figure 7. Calibration curve of the LiDAR sensor. The blue markers represent measured voltage outputs at different distances, while the red curve shows the fitted power-law model $D = 27.16\,V^{-0.714}$. The inverse nonlinear relationship reflects how the sensor produces higher voltages at shorter ranges and lower voltages as distance increases.

AQ1 - Smallest Detectable Object:
As distance increased, larger objects were required for reliable detection. Table I lists the smallest everyday objects that were detected consistently at each distance. At 10-20 cm we could not reliably register objects smaller than about 12×19 mm; by 80 cm, targets around 56×80 mm were required for

Table I
SMALLEST OBJECTS RELIABLY DETECTED AT INCREASING DISTANCES.

Object size (mm)	Max distance (mm)
12 × 19	not measurable
17 × 23	180
22 × 29	300
31 × 40	480
38 × 55	650
56 × 80	800

consistent detection.

AQ2 - Minimum Undetectable Speed:
To estimate the lowest speed at which an object could move without being detected, objects of different sizes were moved laterally across the LiDAR beam at varying distances. At close range (10-20 cm), hand motions of about 0.2-0.3 m/s were sufficient to slip between measurements; at 80 cm, the threshold increased to roughly 0.7 m/s.

AQ3 - Influence of Material Properties:
Matte, opaque targets were detected consistently at the tested sizes and distances, whereas glossy and transparent targets produced intermittent detections, especially at longer ranges.

VII. DISCUSSION

The constructed motor-controlled LiDAR system was able to perform sweeps and generate distance readings with acceptable accuracy, but several limitations became apparent during testing. The calibration confirmed the expected nonlinear relationship between voltage and distance, although deviations increased with range. Within 50 cm, the results were reasonably stable, while at greater distances the readings became noisier and less reliable.

Firstly, for **AQ1** the experiments showed that the smallest detectable object size increased with distance. At 10-20 cm, objects smaller than about 12×19 mm were not detected reliably; at 40 cm, objects around 31×40 mm were required for consistent detection. At 80 cm, targets on the order of 56×80 mm were measured reliably. Since the LiDAR uses infrared light, the large room used for testing, which had windows admitting sunlight, likely raised these thresholds. Conducting measurements in a controlled room without sunlight, or averaging results collected under different environmental conditions, would provide a more representative picture of system performance.

Secondly, for **AQ2** the speed thresholds should be interpreted as approximate because they were measured by moving a hand across the beam and estimating speed from distance and manual timing. Reaction time, uneven trajectories, and the hand's geometry/reflectance affect whether a pass is registered. Daylight in the room likely inflated the speeds needed to avoid detection. More controlled trials (shielded lighting, a linear stage for constant velocity, multiple samples per step or slower sweep/microstepping) would reduce uncertainty and yield tighter, more repeatable thresholds.

Lastly, for **AQ3** the measurements showed that surface properties affected the stability of the distance readings. Matte targets were registered consistently, while glossy or transparent objects produced fluctuating or inconsistent values, and in some cases failed to register at all. Sunlight added background infrared and reduced stability further, so larger dimensions were needed for reliable detection.

In addition to measurement-related factors, hardware and schedule constraints limited the dataset. The plan was to take several repeated runs at each setting and average the results, but the PCB failed early in the second test session. Because the system was built on a compact single-sided PCB and we lacked spare parts and time, creating a new PCB was not feasible within the project window. Consequently, fewer trials were collected than intended, which increases the uncertainty of the reported values. A future revision with a double-sided layout (greater component spacing and better thermal behavior), a replacement sensor, and simple signal filtering or per-step averaging in LabVIEW would enable full replication and more precise, averaged results.

Overall, the system functioned as intended and provided meaningful distance measurements, but its performance was strongly affected by environmental light, object properties, and hardware limitations. These results suggest that while the design works well for basic scanning and short-range object detection, improvements in filtering, PCB design, and controlled testing environments would be necessary to achieve higher reliability.

VIII. CONCLUSION

The prototype LiDAR achieved stable, repeatable sweeps and a calibration consistent with an inverse, nonlinear voltage–distance relation. Using the fitted model $D\,[\text{cm}] = 27.16\,V^{-0.714}$, the system delivered useful accuracy in the near field and acceptable performance to 80 cm. The empirical results show clear trends: for RQ1, the minimum detectable object size grows with range (from not reliable at 12×19 mm near 10-20 cm to about 56×80 mm at 80 cm); for RQ2, the minimum speed required to slip between samples increases with range (about 0.2-0.3 m/s at 10-20 cm and about 0.7 m/s at 80 cm); and for RQ3, surface finish affects stability, with matte, opaque targets performing better than glossy or transparent ones.

Two factors limited the dataset and likely inflated thresholds: daylight in the test room and a mid-project PCB failure that prevented multiple repeated runs and averaging. Even so, the measurements provide credible bounds for object size and motion detection under realistic lab conditions.

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